Richard Tsai

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Profile

An energetic college techie, currently pursuing a BS/MS degree in Computer Science with concentration of ML/AI. Hands-on experiences in installation and configuration of hardware and software including coding and cybersecurity. Like to work on challenging projects, leveraging my skills and knowledge. A self-starter and a team player.

Education | Certifications

Rensselaer Polytechnic Institute **07/2018 – 05/2022**

Bachelor of Science in Computer Science

Courses: DS, Discrete Math, Algo, OS, PSoft, Prog Lang, Data Mining, SD&D

CompTIA Security+ ce: CompTIA License SY0-501 **08/2019 – 08/2022**

Experience

Cybersecurity Defensive Operations Programming Intern **05/2021 – 08/2021**

The MITRE Corporation

- Code development in agile-based scrum team to deliver feature sets for sponsor
- Performed as a full-stack developer in an IDS/IPS project with Postgres and Redis, Python flask, and VueJS
- Leveraged JIRA for project tasks, assignments & collaboration with team members
- Performed validations and determine accuracy of API endpoint by testing and evaluating via pytest framework

Cybersecurity Defensive Operations Intern **06/2020 – 08/2020**

The MITRE Corporation

- Performed analysis of network & host-based log data using Zeek & companioned with ELK dashboard
- Monitored and reviewed application that crawls websites
- Developed custom admin dashboard and reporting using AngularJS
- Participated in team discussions, collaboration, and coding assignments

Helpdesk Student Consultant **07/2019 – Present**

Rensselaer Polytechnic Institute

- Performing computer troubleshooting for university community
- Troubleshooting printer problems throughout campus, performing LMS maintenance, & diagnosing student laptop problems
- Providing support to students & faculties via phone call and remote desktop via Bomgar
- Monitoring incoming issues service tickets, track issues, and troubleshooting issues via Zendesk software

Personal Projects

YACS – Open-Source Schedule Management @ RCOS – Project Leader

Yet Another Course Scheduler – an open-source project to help students plan out their semester

- Introduced and ramped up new developers into the team and managed team in an agile based environment
- Seek new issues caught on staging or production environments and track feature requests made by the public
- Maintain and beefing security on both staging server on Oracle Cloud Infrastructure and on-premises production server

Lil Tim

Another Discord Bot

- Developed in Java using the Java Discord API (JDA), Lavaplayer library for music, and others for different features
- Configured an automated CI/CD solution to bring updates from master branch to on-premises raspberry pi server
- Deployed via GitHub Workflow and Docker containers, built in Gradle build tool

Personal Home Lab

On-going personal home lab to add more future projects/servers

- Monitoring and securing network traffic with FortiGate NGFW with custom IPv4 policies, DoS policies and VPN
- Configured port-forwarded on-premises file share server on raspberry pi with necessary firewall settings
- Use to try different technologies and acquire new skills and ways for deploying servers

Skills

- *Programming Languages:* C, C++, Java, Python, SQL, Git, Node.JS frameworks
- *Operating Systems / Platforms:* Windows, macOS, Linux, Amazon Web Service, Docker, Kubernetes, GitHub, GitLab